

Homemade XR: Enriching conversations and shortening social distancing

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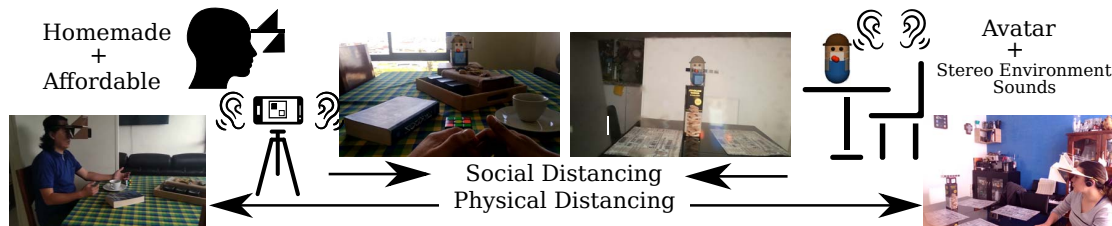


Figure 1: Sketch of concept. A homemade extended reality system with multimodal cues (visual and auditory), to enrich conversations and shorten social distancing generate sense of proximity because “physical distancing” is not synonym of “social distancing”.

ABSTRACT

This work shows an alternative way to shortening social distancing without breaking physical distancing, a medical recommendation in some health conditions, as the current COVID-19 pandemic. As an alternative to traditional video conferencing, this proposal allows persons to interact with their natural space while actively participating in a remote call. Each participant’s reality is extended with a visual representation of the other and binaural sound, enriching the conversation by the sense of the other’s proximity. This proposal uses quotidian technology, i.e., smartphones, and an immersive homemade device, to provide an affordable system to display and capture audio stereo and video.

Index Terms: Human-centered computing—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Interaction paradigms;

1 INTRODUCTION

The SARS-CoV-2 virus, which causes the COVID-19 disease, is transmitted through the respiratory system’s particles. The countries have applied various techniques to avoid a high spread factor of this virus: requiring the use of a mask, confinements, quarantines, and establishing a minimum interpersonal physical distance of at least 1.5m to 2m. However, confinement causes a significant impact on a psychological and psychosocial level. Lengthy quarantines produce fear, frustration, boredom. Even so, medical associations worldwide agree that it remains a fundamental strategy to mitigate the spread of the virus and thus avoid the collapse of public health systems [1]. For this reason, it is essential to promote strategies that strengthen the mental health of people who stay at home.

In this work, we ask how to use quotidian technologies to extend other forms of reality that reduce social distance while maintaining a physical distance. Thus, our proposal is a homemade extended reality (XR) system to enrich conversations and shorten social distancing without breaking preventive self-isolation. This is an alternative to traditional videoconference that allows persons to interact with their natural spaces while actively participating in a remote call. In this way, each participant’s reality is extended with multimodal cues (visual and auditory) [6] that generate feelings of closeness and

proximity because “physical distancing” is not synonym of “social distancing”. This paper is organized as follows: section 2 presents selected related work and key concepts related to our proposal. Section 3 explains our approach, how is its architecture, philosophy, and implementation; finally, section 4 contains the discussion and final remarks.

2 BACKGROUND AND RELATED WORK

Social interaction mediated by virtual environments has been a constant motivation in these topics. For example, early studies in collaborative virtual environments highlighted the relevance of generating adequate co-presence levels through awareness strategies (cues), including avatar features, work status feedback and proximity detection [3]. The current social interaction offer in virtual spaces includes new systems as Mozilla Hubs and Facebook Horizon, even classic Second Life, where the interaction is mediated by 3D avatars [4]. Other approaches include augmented, mixed, and extended reality with options as a realistic avatar, 3D reconstruction of the real environment, or collaborative working space; even there are some commercial offers as Spatial and Cisco EVAR. Still, these proposals require specialized equipment that is not affordable by an end-user [5].

3 OUR APPROACH

In this work, we propose a strategy to reduce social distance while maintaining a physical separation. To do this, we resort to the concept of proxemics, through digital mediation, to enrich video calls with family and friends. Thus, the purpose is that the person interacting in the call should not be seated and do not restrict interaction to a single screen. Instead, the interlocutors can actively participate in the video call while doing other activities, for example, preparing a cooking recipe remotely, walking, and doing activities with physical objects, such as handicrafts. This strategy also uses low-cost and easy-access digital technologies to extend reality by seeking that a broader range of people can do it at home; hence, the adjective homemade. During the global COVID-19 pandemic, digital technologies to mediate communication between family and friends, support remote work, or maintain educational activities have been essential. In particular, video calls have been extended, which has involved new forms of proxemics, such as how interpersonal communications are, how we conceive the physical space and the body of others while the communication takes place [2]. The screen has become the most common means of “being with others”, in two dimensions, with an opening angle restricted by the camera, and with background spaces that are usually blurred or hidden. However, this “being with

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others” through the screen has limited how we inhabit our domestic spaces and restricts how we act in our territory (home) with our stuff. This proposal inquires this way of making video calls as an alternative way of holding a conversation while maintaining and sharing a relationship with the physical space we inhabit. On the one hand, from the visual, using an avatar representing the other augments the physical reality of an environment in our house, and the auditory, capturing in stereo the sounds surrounding the space. Thus, from an embodied perspective allowing the person who talks to walk, move, and interact with objects or people.

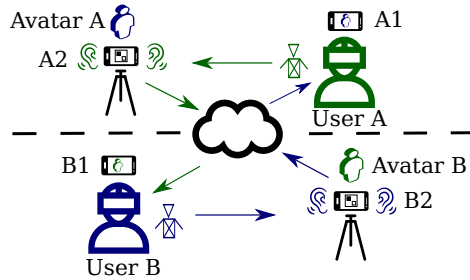


Figure 2: General diagram of the system architecture.

Each user requires two mobile devices with Android OS. Figure 2 shows a general diagram of the system architecture. The smartphone A2 is the capture device for the user A. On the one hand, it records the ambient sound and voice of the user A in a binaural mode, using the two microphones placed in the top and bottom of the device, considering that the microphones' distance is similar to our ears. On the other hand, this device tracks head and hand movement of the user A. The head and hand tracking are implemented using a Postnet library, but we only tracks 5 points of interest: the two shoulders, the two wrists and any point of the face. This information is transmitted using audio streaming and JSON packets by a socket server developed for this purpose. The second smartphone, for the explanation, B1 is the display device. It is embedded in a cardboard real mixed head-mounted display (HMD). The smartphone screen renders a stereo image that is projected in the two way mirror surface and placed over the an augmented reality marker; this is possible because the mirror in the top of the cardboard HMD allows the camera tracking. This device, B1 reproduces stereo sound in similar conditions as were recorded, and projects a visual representation (avatar) of user A. The same process is carried out in the other way. The capture device B2 records the sound in binaural mode, and tracks head and hand movement of the user B, and the display device A1 reproduces stereo sound and projects the avatar of user B.

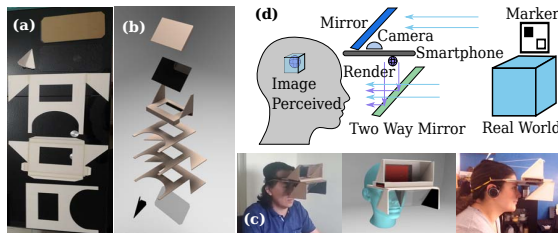


Figure 3: Detail of the mixed reality cardboard HMD

Figure 3 shows the detail of the mixed reality cardboard HMD. It is possible to see in (a) a photo of cardboard pieces before assembly and in (b) a diagram of the assembly process. The material of assembled are cardboard, two way mirror film and a mirror. (c) shows a user with the HMD, and (d) a functional schema. Two-way

mirror allows a user to see the real environment simultaneously to smartphone screen reflection. A stereo image is rendered over a black canvas, and the sheet between eyes allows that user to see right and left images independently. The mirror in the top enables using the back camera to track AR markers. Assembly instruction and blueprint are available to public download¹.

4 DISCUSSION AND FINAL REMARKS

Our approach allows users to be-in-their space and extend their reality with the other's avatar and stereo sounds. In this way, using a combination of visual and auditory cues, it is possible to generate a sense of proximity enriching users conversation. This approach is an alternative to traditional video-conference that allows persons to interact with their natural spaces while actively participating in a remote call. Additionally, specialized equipment is not required; the users only need common materials and quotidian technology, which is affordable and inclusive. Internal assessments showed our approach's technical viability as functional and fun, and the users enjoyed assembling the HMD and setting up the physical space. Although it is uncertain what will happen post-pandemic, even now, starting in 2021, the strategy of staying at home continues to be predominant, hand in hand with the need for people to voluntarily comply with this measure, maintaining their mental and physical health. This situation allows us to think that there is a potential development of research and innovations around alternatives to maintain physical distance while reducing social distance. In particular, in this work, we find it essential to continue ergonomics research, to provide an improved, more comfortable, and dynamic HDM model that adapts to different cell phones and diverse heads, to wear for long periods. Also, anthropological and communicative research around the concept of "space". Here, the spatial is conceived in a hybrid way, mixing physical and virtual-the physical, like what we inhabit, like our home; the virtual, through the avatar representing the person with whom we talk. It is also necessary to do an in-depth and rigorous evaluation of how this hybrid space of the proposal videocall system modifies interpersonal communications from a proxemic perspective.

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¹<https://sites.google.com/unimilitar.edu.co/gim-researchlab/downloads/ieecvr2021-3dui>